



DS-001 - Introdução à análise de dados com Python

Prof. Herbert Rocha

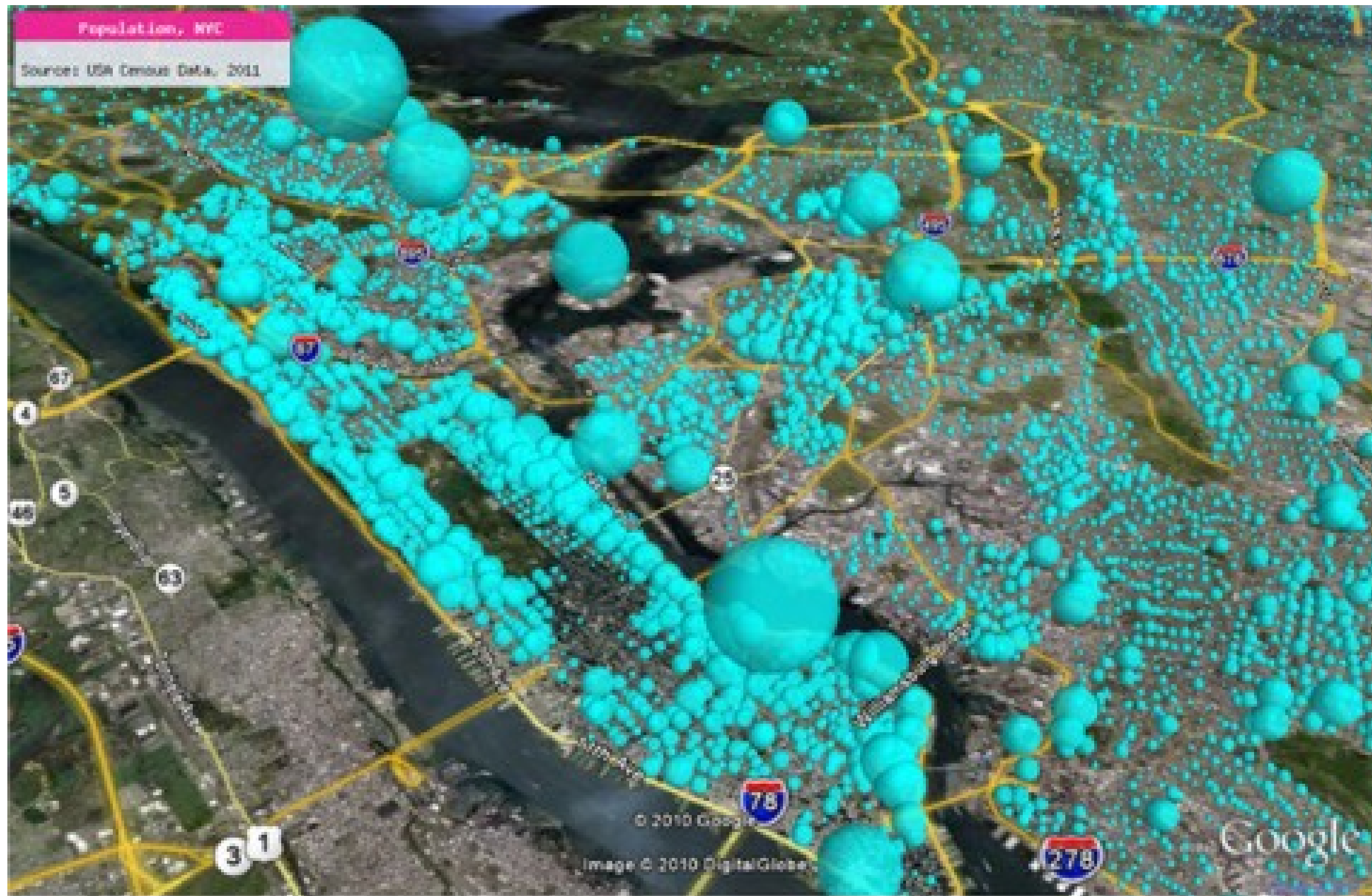
Aula 02 - Manipulações Geometricas



Explorando à Geovisualização

- Falando mais sobre geobr
- Tratando dataset
- Agrupamentos de dados
- Manipulação geométrica
- Entendendo latitude e longitude
- CRS
- Plot de pontos no Brasil

Análise Espacial de Dados



General usage

Available data sets

The geobr package covers 27 spatial data sets, including a variety of political-administrative and statistical areas used in Brazil. You can view what data sets are available using the `list_geobr()` function.

```
# Available data sets  
geobr.list_geobr()
```

```
Function: read_country  
Geographies available: Country  
Years available: 1872, 1900, 1911, 1920, 1933, 1940, 1950, 1960, 1970, 1980, 1991, 2000, 2001, 2010, 2013, 2014, 2015, 2016, 2017, 2018, 2019  
Source: IBGE  
-----  
Function: read_region  
Geographies available: Region  
Years available: 2000, 2001, 2010, 2013, 2014, 2015, 2016, 2017, 2018, 2019  
Source: IBGE  
-----  
Function: read_state  
Geographies available: States  
Years available: 1872, 1900, 1911, 1920, 1933, 1940, 1950, 1960, 1970, 1980, 1991, 2000, 2001, 2010, 2013, 2014, 2015, 2016, 2017, 2018, 2019  
Source: IBGE  
-----
```

<https://www.ibge.gov.br/geociencias/organizacao-do-territorio/malhas-territoriais/15774-malhas.html>

≡ Instituto Brasileiro de Geografia e Estatística

Buscar



Malha Municipal

O que é

Edições



2022



Sobre a publicação

Acesso ao produto

Acesso ao produto - 2022

Brasil

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WebMapa: Desenvolvimento de Mapas Web por um Complemento QGIS

WebMap: Web Map Development by a QGIS Complement

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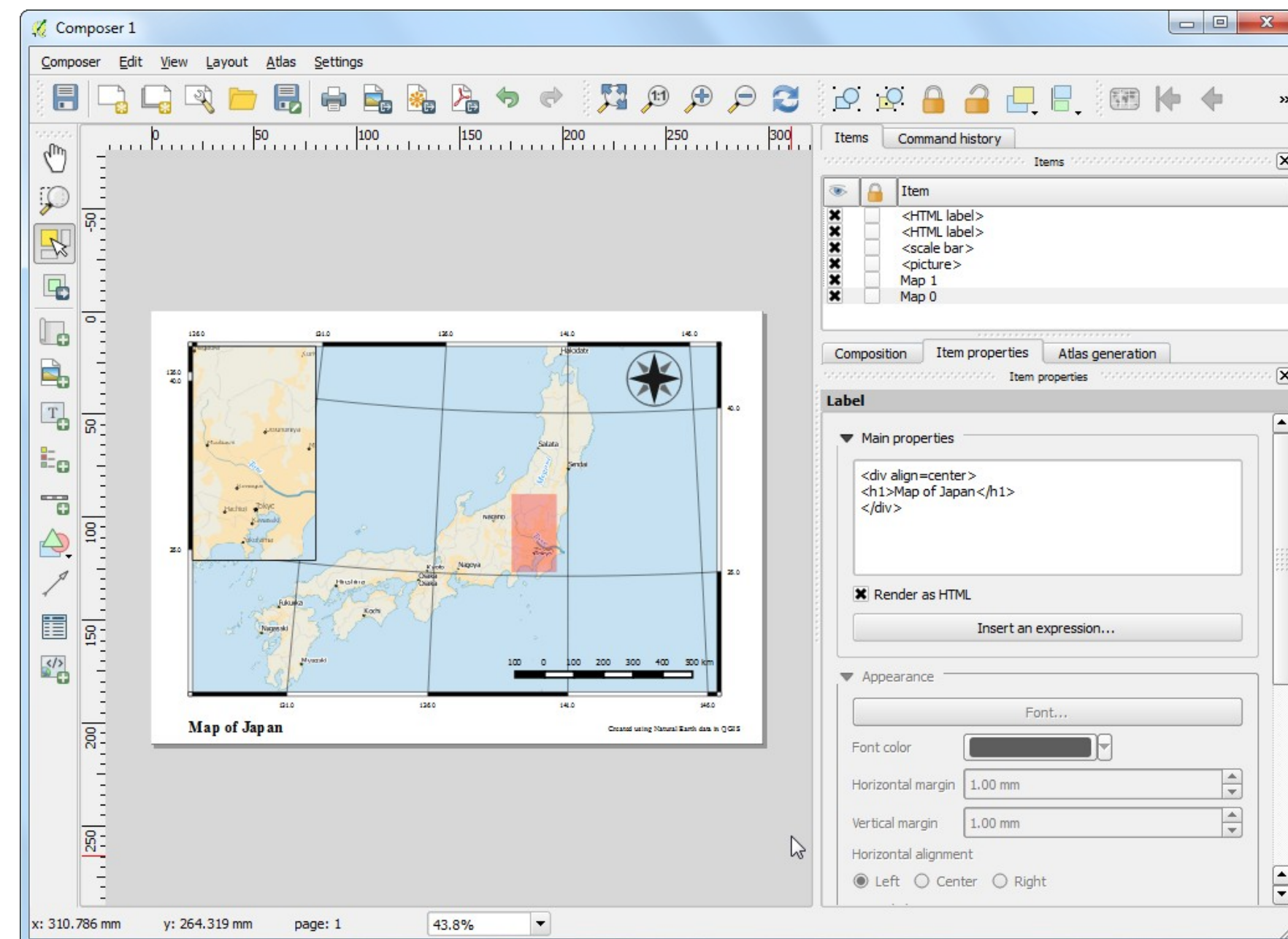
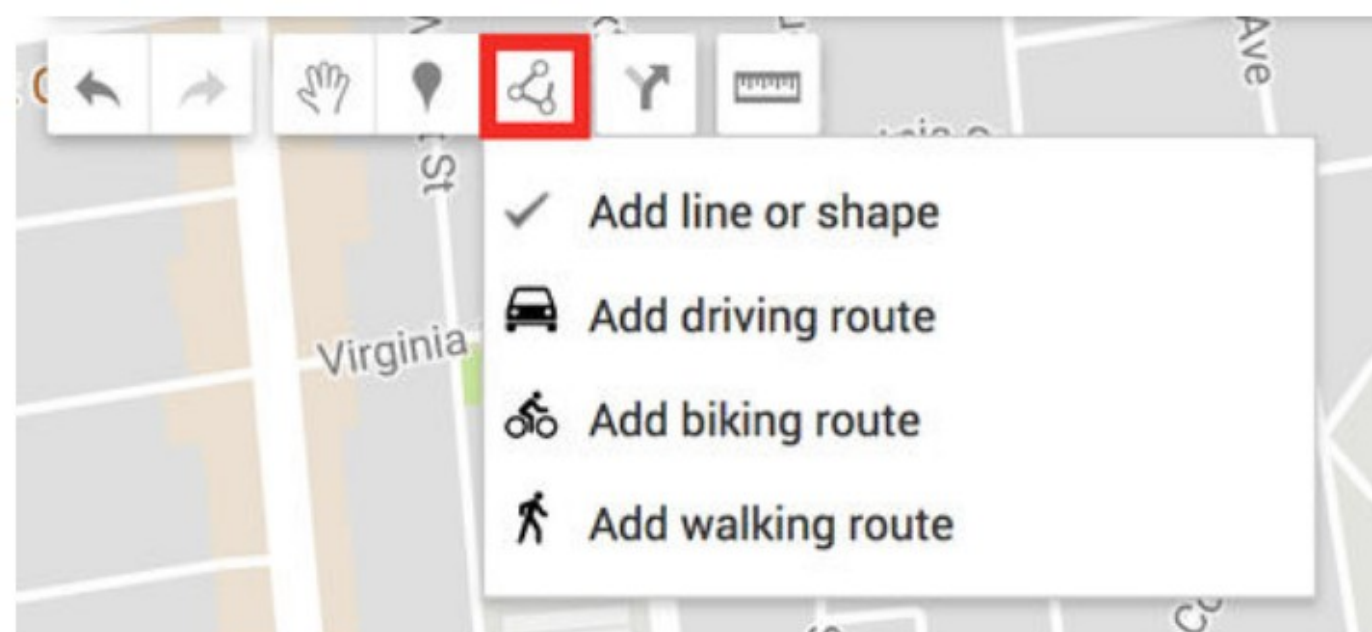
Figura 4 – Camadas vetoriais utilizadas para a validação dos resultados.



Fonte: CampusMap, 2020.

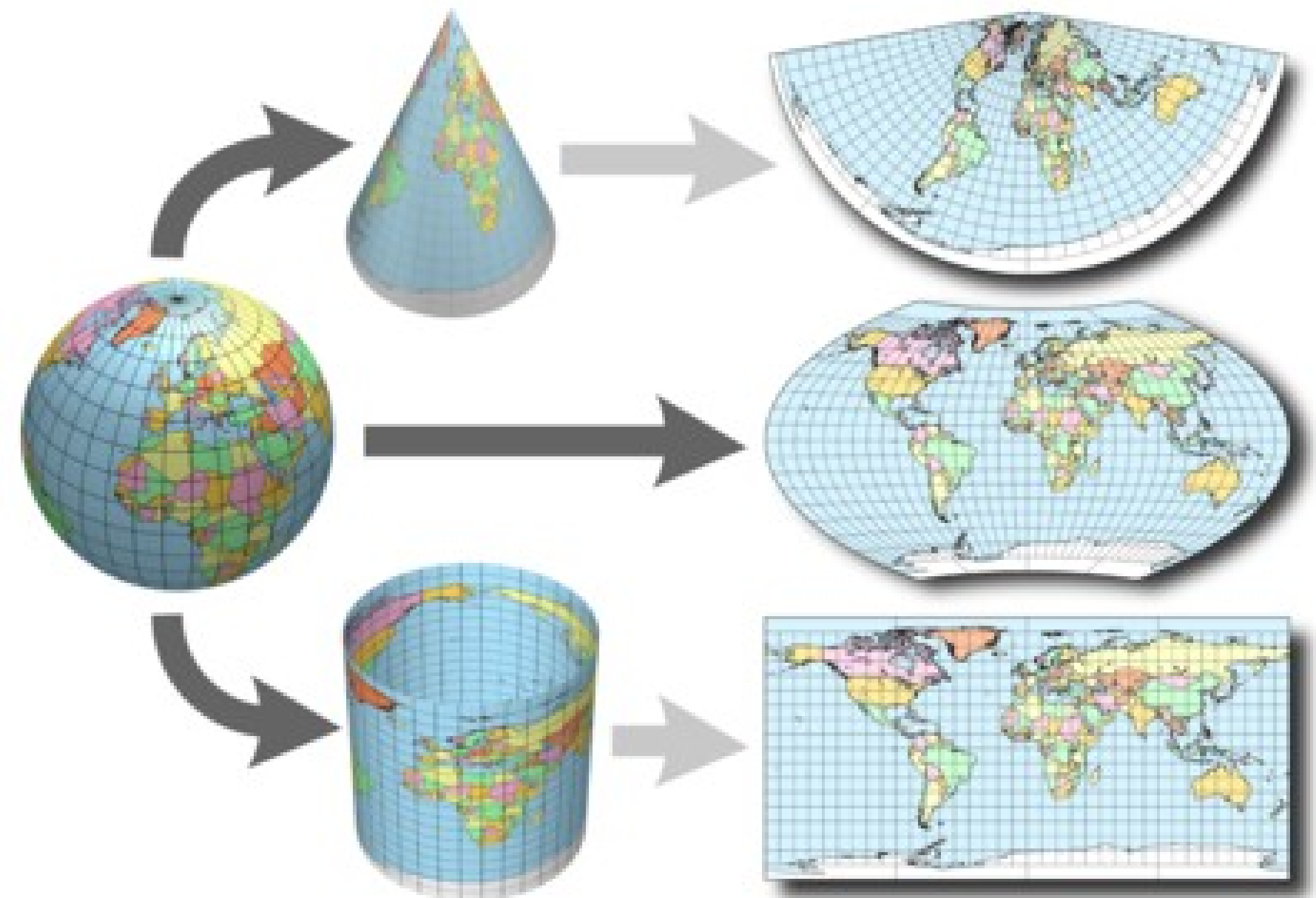
Create Lines and Shapes

You can also create lines or shapes on your map as well by clicking the “Draw a Line” tool. You’ll see several options from the dropdown menu, including ones for creating driving, biking, and walking routes. For the moment, let’s just use the “Add a line or shape” tool.



Análise Espacial de Dados

- Verificando o sistema de coordenadas
- Lidando com valores ausentes
- Detectando e removendo outliers
- Padronizando e normalizando os dados
- Validando e verificando os dados



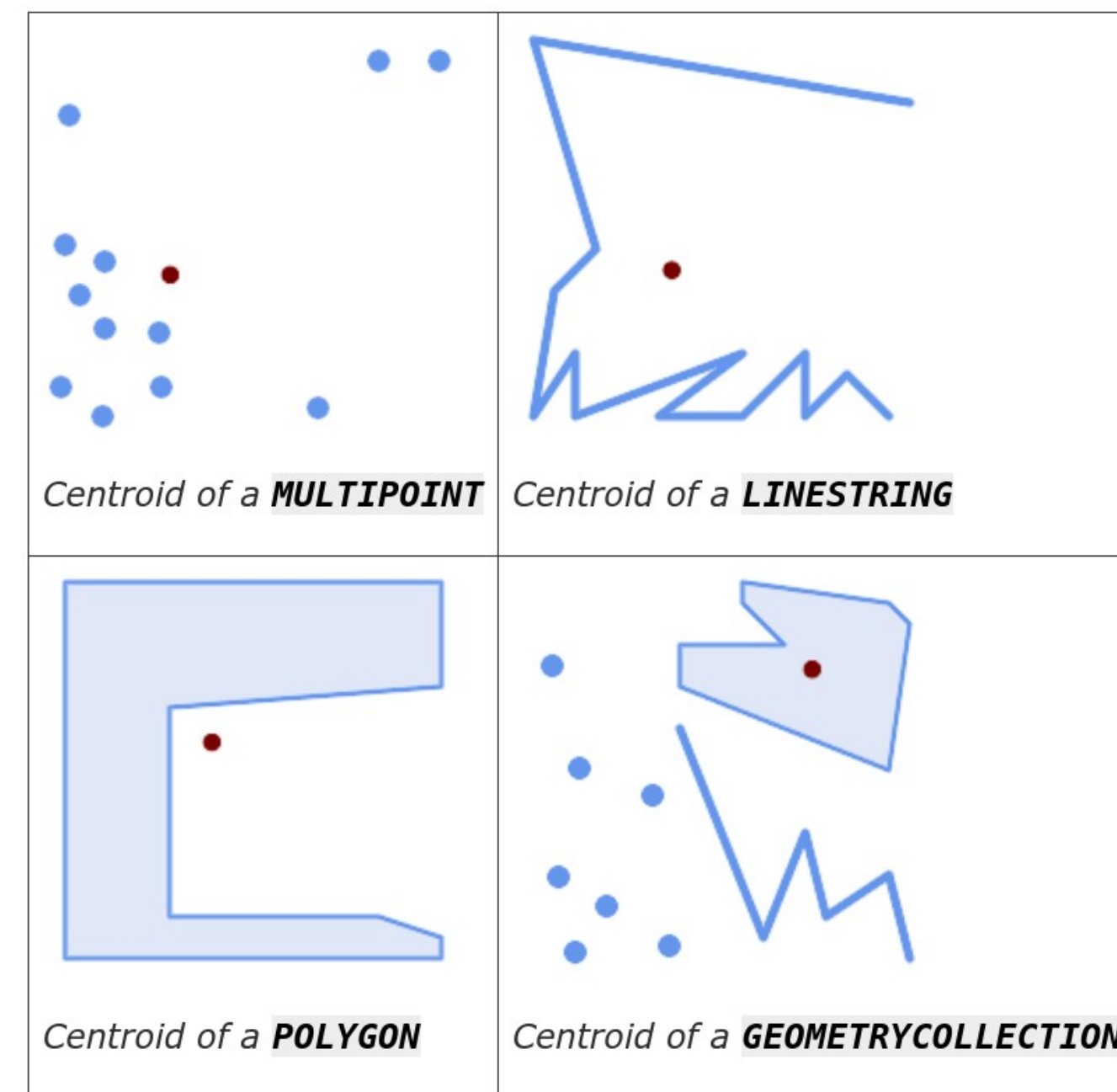
Geometric Centroid

The centroid is center of mass of a two-dimensional planar **lamina** or a three-dimensional solid. The mass of a **lamina** with surface density function $\sigma(x, y)$ is

$$M = \iint \sigma(x, y) dA,$$

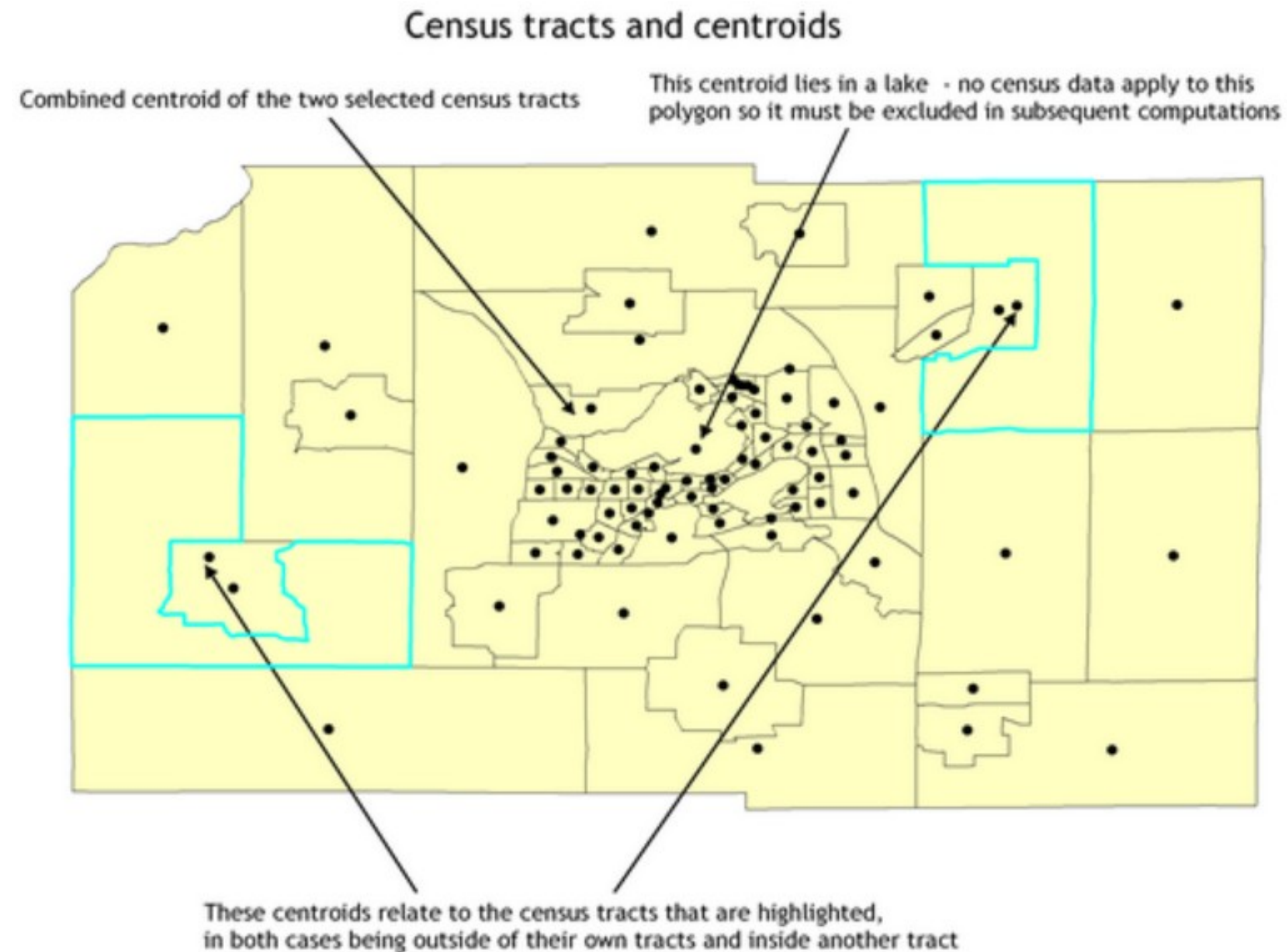
and the coordinates of the centroid (also called the center of gravity) are

$$\bar{x} = \frac{\int x \sigma(x, y) dA}{M}$$
$$\bar{y} = \frac{\int y \sigma(x, y) dA}{M}.$$



<https://mathworld.wolfram.com/GeometricCentroid.html>

https://postgis.net/docs/ST_Centroid.html

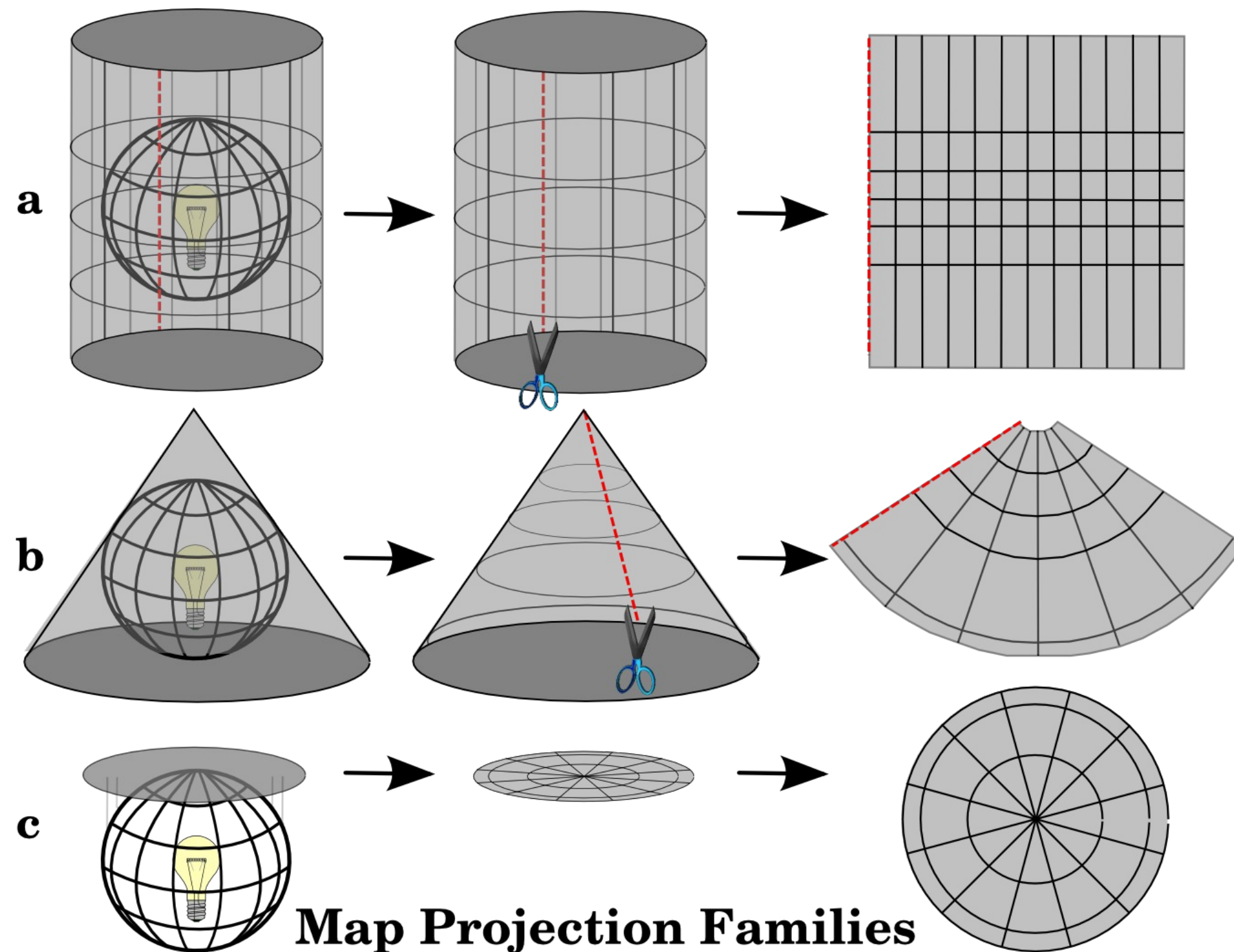


https://www.spatialanalysisonline.com/HTML/centroids_and_centers.htm

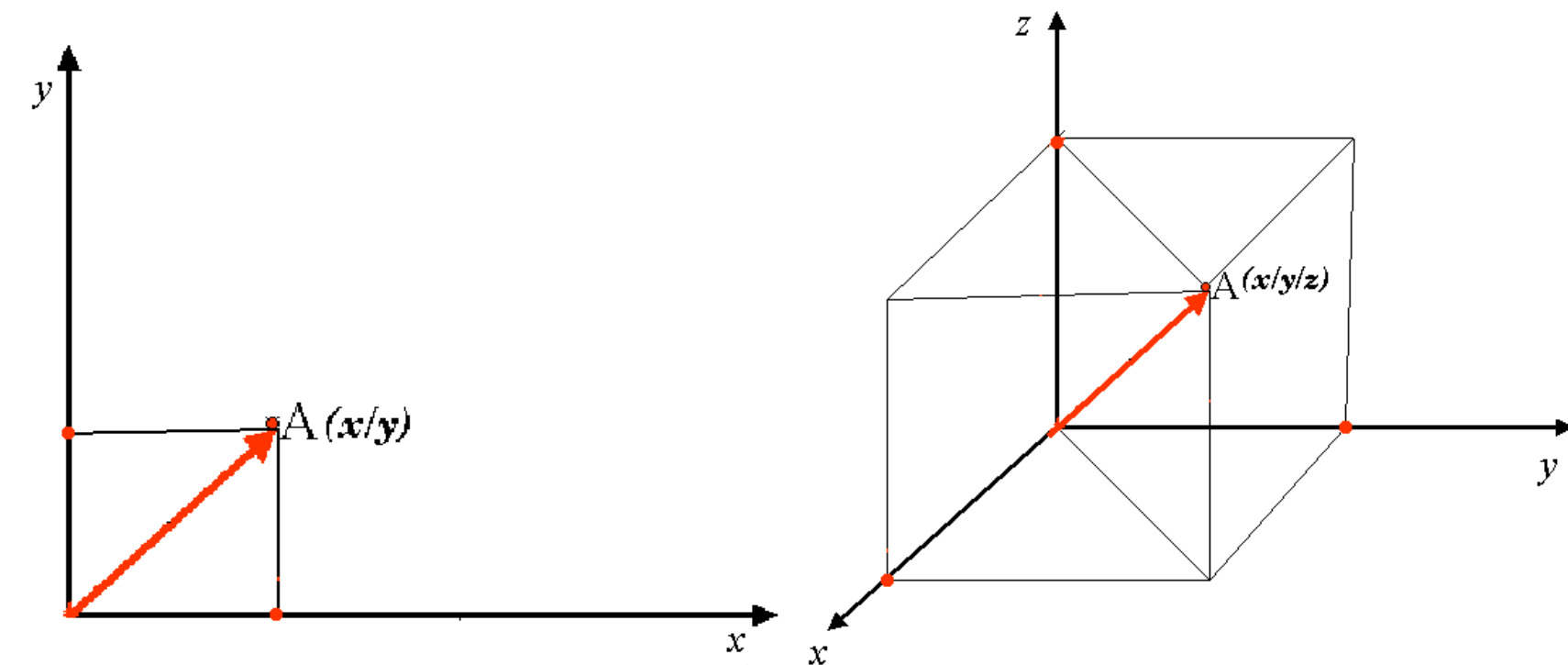
Análise Espacial de Dados

Municípios do Acre	Códigos
A Acrelândia	1200013
Assis Brasil	1200054
B Brasiléia	1200104
Bujari	1200138
C Capixaba	1200179
Cruzeiro do Sul	1200203
E Epitaciolândia	1200252
F Feijó	1200302
J Jordão	1200328
M Mâncio Lima	1200336
Manoel Urbano	1200344
Marechal Thaumaturgo	1200351
P Plácido de Castro	1200385
Porto Acre	1200807
Porto Walter	1200393
R Rio Branco	1200401
Rodrigues Alves	1200427

<https://www.ibge.gov.br/explica/codigos-dos-municipios.php>



Geographic Coordinate Systems WGS 84



https://docs.qgis.org/3.34/en/docs/gentle_gis_introduction/coordinate_reference_systems.html

Análise Espacial de Dados

Um sistema de referência de coordenadas projetado no **hemisfério sul (ao sul do equador)** normalmente tem sua **origem no equador** em uma longitude específica. Isso significa que:

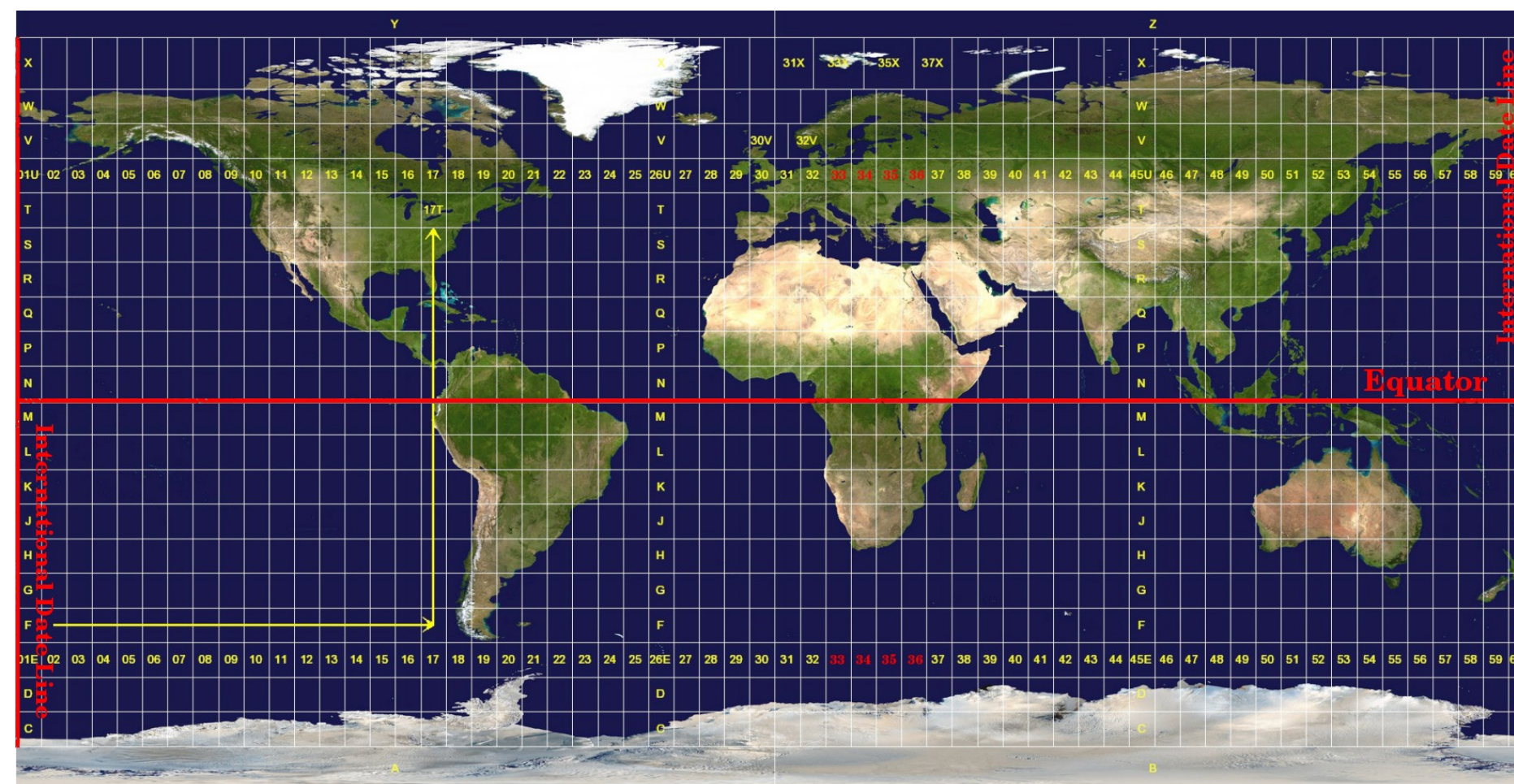
- os valores Y aumentam para o sul
- os valores X aumentam para o oeste.

No **hemisfério norte (ao norte do equador)**, a origem também é o equador em uma longitude específica. Isso significa que:

- os valores Y aumentam para o norte
- os valores X aumentam para o leste.

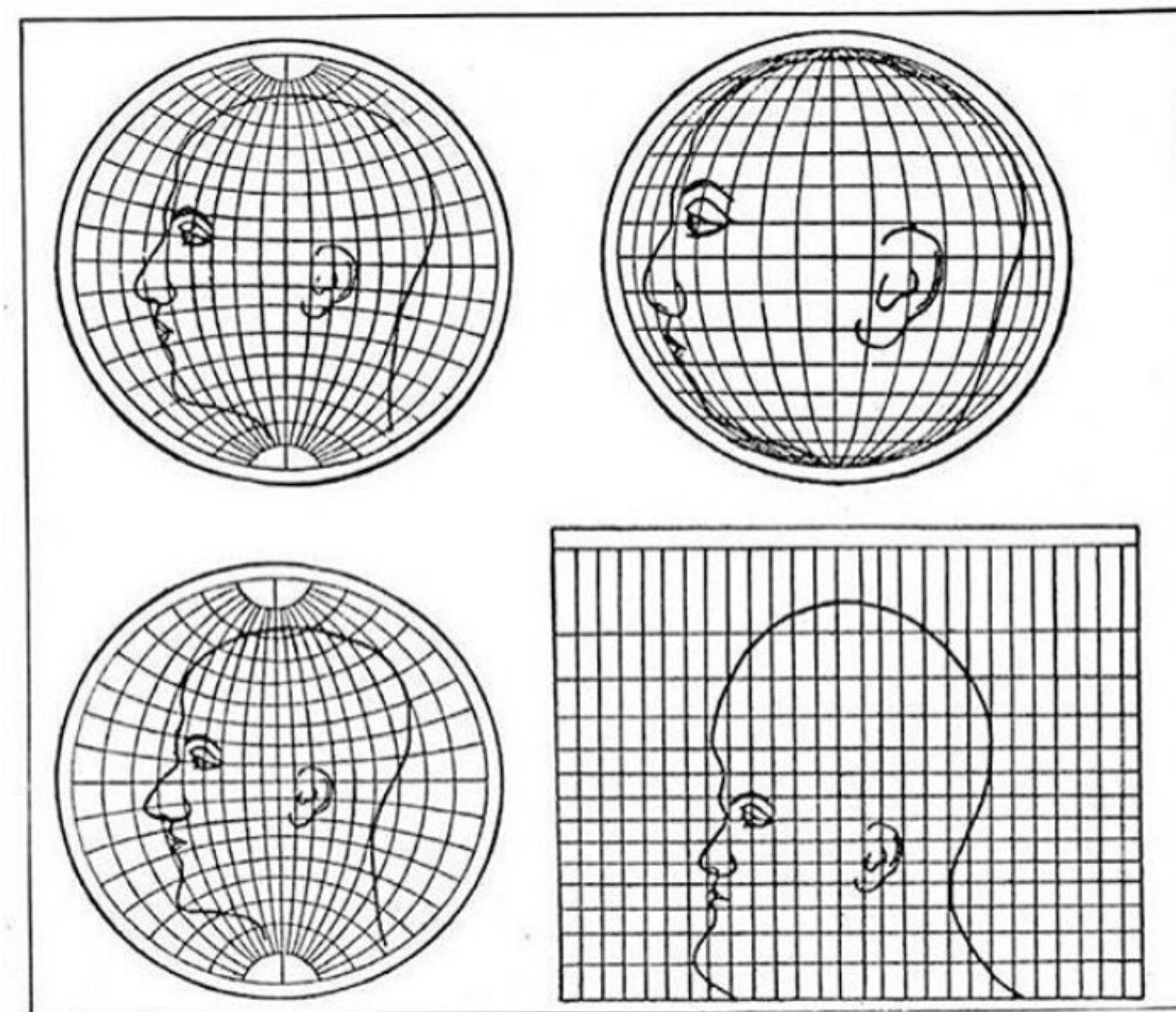
Exemplo:

Transverse Mercator (**UTM**),
frequentemente usado para a África do Sul.



Análise Espacial de Dados

<https://www.earthdatascience.org/courses/earth-analytics/spatial-data-r/intro-to-coordinate-reference-systems/>

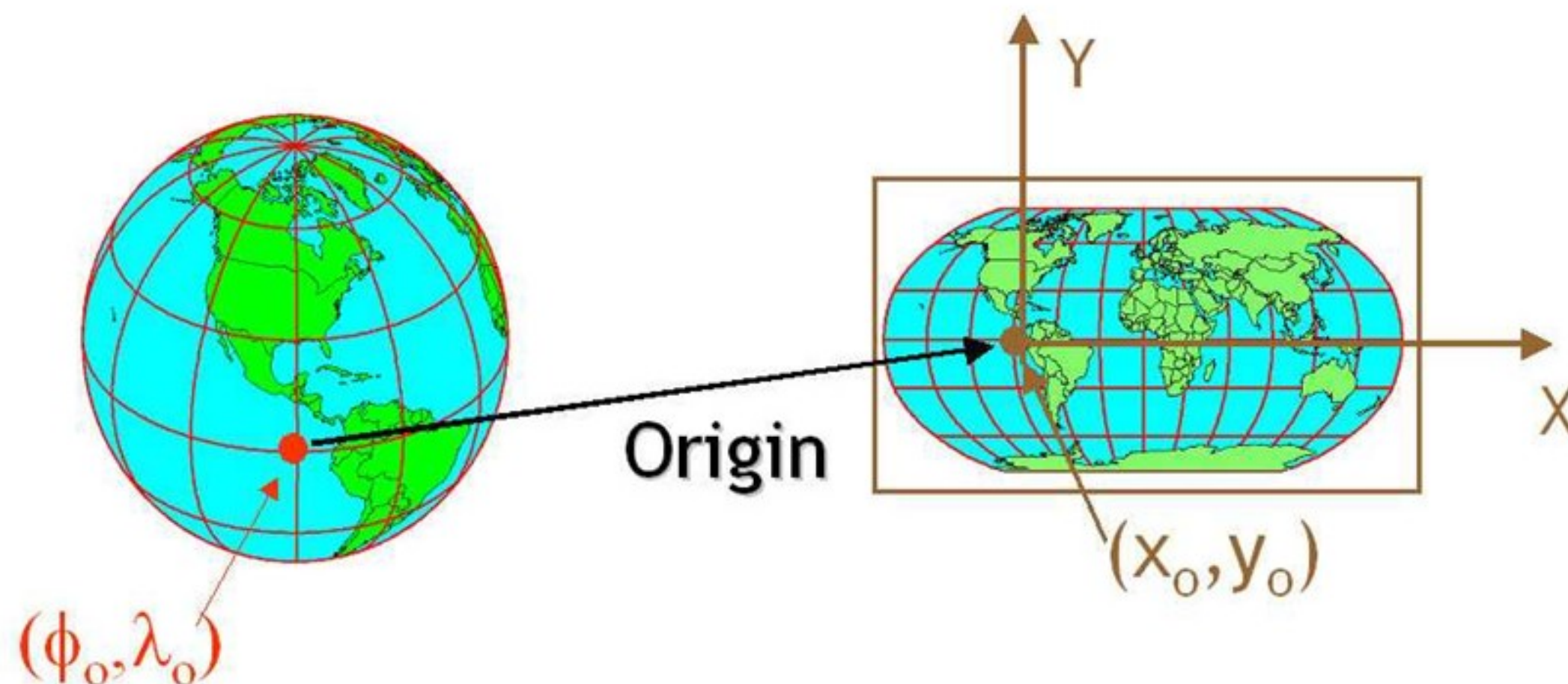


*Upper left: Globular. Upper right: Orthographic. Lower left: Stereographic.
Lower right: Mercator*

What four commonly used projections do, as shown on a human head

<https://www.earthdatascience.org/courses/earth-analytics/spatial-data-r/intro-to-coordinate-reference-systems/>

Enquanto o sistema de coordenadas é **bidimensional**, nós vivemos em uma Terra **tridimensional** que por acaso é “redonda”. Para definir a localização de objetos na Terra, que é redonda, você precisa de um sistema de coordenadas que se adapte ao formato da Terra. Quando você faz mapas em papel ou em uma tela plana de computador, você se move de um espaço tridimensional (o globo) para um espaço bidimensional (as telas do seu computador ou um pedaço de papel).



EPSG 3857 or 4326 for Web Mapping

Asked 11 years, 7 months ago Modified 1 year, 5 months ago Viewed 284k times



The discussion at [What is the difference between WGS84 and EPSG4326?](#) shows that 4326 is just the EPSG identifier of WGS84..

250

Wikipedia entries for Google Maps and OpenStreetMap shows that they both use WGS 84.



<http://wiki.openstreetmap.org/wiki/EPSG:3857> states that



EPSG:3857 is a Spherical Mercator projection coordinate system popularized by web services such as Google and later OpenStreetMap.



Leaflet's help states:

EPSG3857 The most common CRS for online maps, used by almost all free and commercial tile providers. Uses Spherical Mercator projection. Set in by default in Map's crs option.

EPSG4326 A common CRS among GIS enthusiasts. Uses simple Equirectangular projection.

This is confusing - it seems that Google Maps and OpenStreetMap use EPSG3857 but they use WGS84 which 'is' EPSG4326. Something can't be right here, most likely my understanding.

Could someone help me understand?

EPSG:3857

WGS 84 / Pseudo-Mercator -- Spherical Mercator,
Google Maps, OpenStreetMap, Bing, ArcGIS, ESRI

This is projected coordinate system used for rendering maps in Google Maps, OpenStreetMap, etc. For details see Tile system.

Alternatives codes : 900913 3587 54004 41001 102113 102100 3785

Attributes

Unit: metre

Geodetic CRS: WGS 84

Datum: World Geodetic System 1984 ensemble

Data source: EPSG

Information source: Microsoft.

Revision date: 2020-03-14

Scope: Web mapping and visualisation.

Remarks: Not a recognised geodetic system. Uses spherical development of ellipsoidal coordinates. Relative to WGS 84 / World Mercator (CRS code 3395) gives errors of 0.7 percent in scale and differences in northing of up to 43km in the map (21km on the ground).

Area of use: World between 85.06°S and 85.06°N.

Share on:



Covered area powered by MapTiler



Center coordinates

0.0 0.0

Projected bounds:

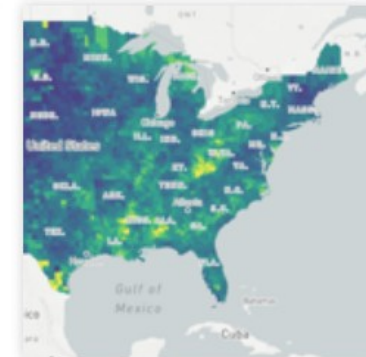
-20037508.34 -20048966.1

Análise Espacial de Dados

<https://plotly.com/python/maps/>



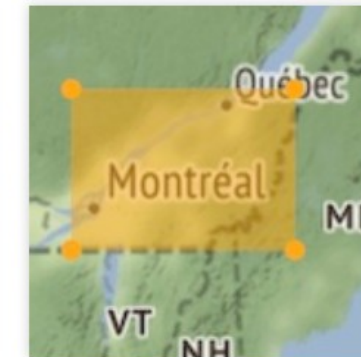
MapLibre Migration



Tile Choropleth Maps



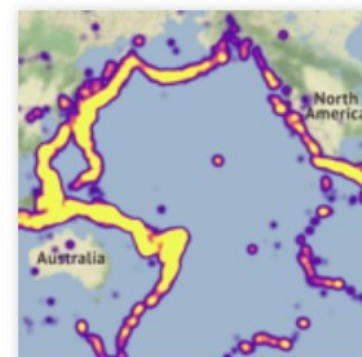
Lines on Tile Maps



Filled Area on Tile Maps



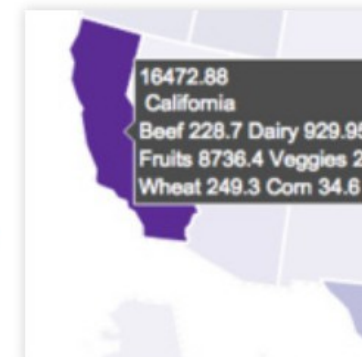
Bubble Maps



Density Heatmap



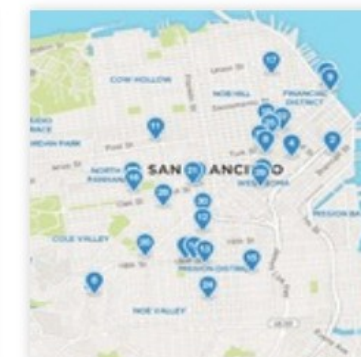
Lines on Maps



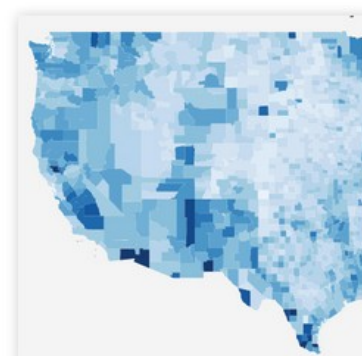
Choropleth Maps



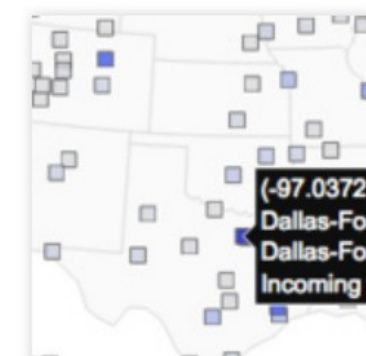
Tile Map Layers



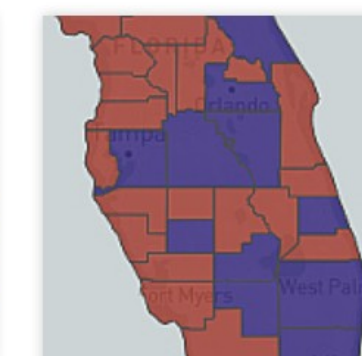
Scatter Plots on Tile Maps



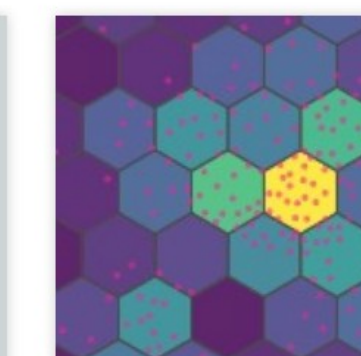
USA County Choropleth Maps



Scatter Plots on Maps



Map Configuration and Styling on Geo Maps

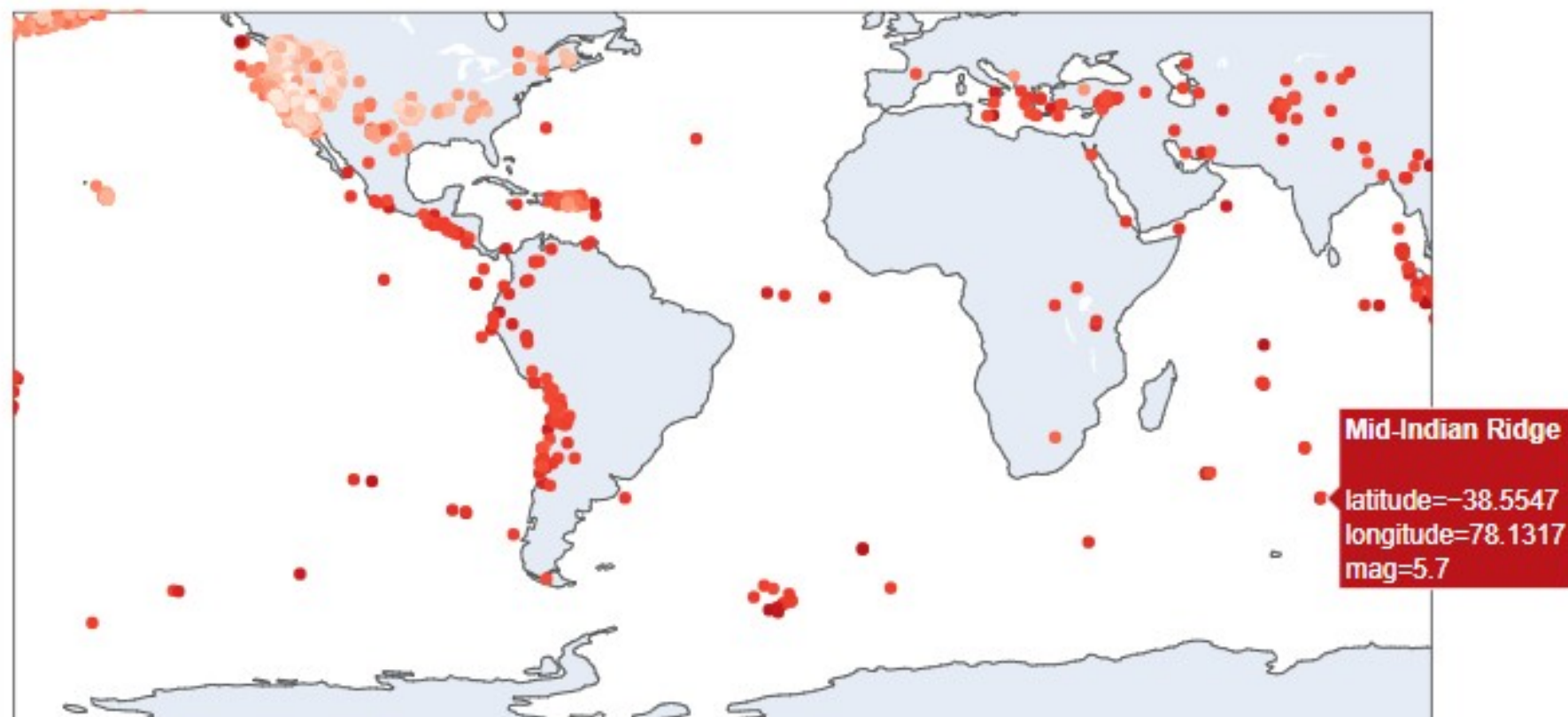


Hexbin Mapbox



Análise Espacial de Dados

```
fig = px.scatter_geo(data, lat='latitude', lon='longitude', color='mag',  
                    hover_name='place', color_continuous_scale='reds',  
                    title='Earthquakes Around the World')  
  
fig.show()
```



<https://www.datacamp.com/tutorial/making-map-in-python-using-plotly-library-guide>

ATÉ A PRÓXIMA AULA!

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